Beantwoord vrae 1 tot 13 op die MERKLESERVORM se KANT 1 /
Answer questions 1 to 13 on the OPTIC READER FORM on SIDE 1

Vraag 1 / *Question* 1

Bepaal / *Find*: $\lim_{x \to \frac{\pi}{2}} (1 - \cos x)$

(1 *a*) $1 - \frac{1}{\sqrt{2}}$	(1 *b*) 0	(1 *c*) die limiet bestaan nie / *the limit does not exist*	(1 *d*) 1

Ans: $\lim_{x \to \frac{\pi}{2}} (1 - \cos x) = \lim_{x \to \frac{\pi}{2}} 1 - \lim_{x \to \frac{\pi}{2}} (\cos x) = 1 - 0 = 1$
$\left(\lim_{x \to a}[f(x) - g(x)] = \lim_{x \to a} f(x) - \lim_{x \to a} g(x)\right)$

(1 *d*)

Vraag 2 / *Question* 2

Bepaal / *Find*: $\lim_{x \to 0} \frac{\pi}{4}$

(2 *a*) die limiet bestaan nie / *the limit does not exist*	(2 *b*) 0	(2 *c*) $\frac{\pi}{4}$	(2 *d*) 45^0

Ans: $\lim_{x \to 0} \frac{\pi}{4} = \frac{\pi}{4}$ (limiet van 'n konstante / *limit of a constant*)

(2 *c*)

Vraag 3 / *Question* 3

Bepaal / *Find*: $\lim_{x \to 0^+} \left(\ln \frac{1}{x}\right)$

(3 *a*) ∞	(3 *b*) die limiet bestaan nie / *the limit does not exist*	(3 *c*) $-\infty$	(3 *d*) 1

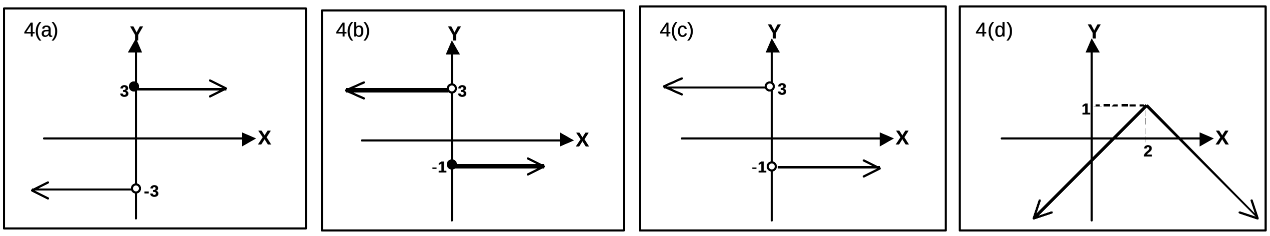
Ans: $\lim_{x \to 0^+} \left(\frac{1}{x}\right) = +\infty \therefore \lim_{x \to 0^+} \left(\ln \frac{1}{x}\right) = \infty$ (uit die grafiek van / *from the graph of* $y = \ln x$)

(3 *a*)

Vraag 4 / *Question* 4

Laat / *Let* $f(x) = \frac{x - 2|x|}{x}$.

Die grafiek van f word gegee in... / *The graph of f is given by...*



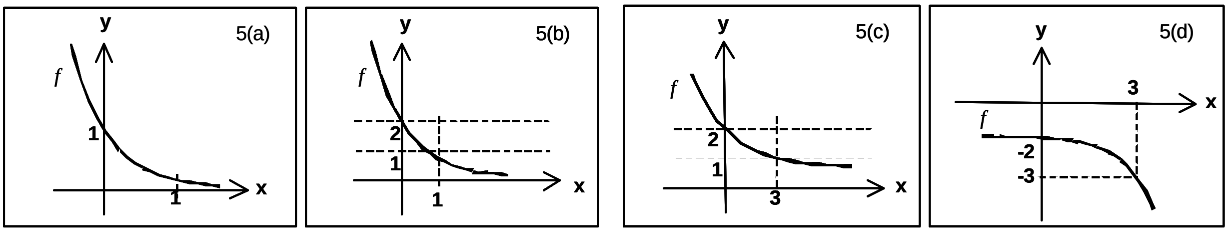
Ans: $f(x) = \frac{x - 2|x|}{x} = \begin{cases} \frac{x - 2(x)}{x} & \text{as / if } x > 0 \\ \frac{x - 2(-x)}{x} & \text{as / if } x < 0 \end{cases} = \begin{cases} -\frac{x}{x} & \text{as / if } x > 0 \\ \frac{3x}{x} & \text{as / if } x < 0 \end{cases} = \begin{cases} -1 & \text{as / if } x > 0 \\ 3 & \text{as / if } x < 0 \end{cases}$

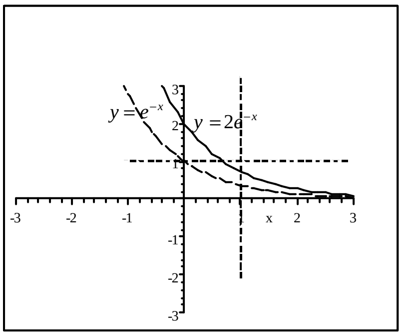
(4 *c*)

Vraag 5 / *Question* 5

Laat / *Let* $f(x) = 2e^{-x}$.

Die grafiek van f word geskets in... / *The graph of* f *is given by...*





Ans: $f(x) = 2e^{-x}$

$(5\,b)$

Vraag 6 / *Question* 6

Vereenvoudig / *Simplify*: $\sqrt{x^2 - 2x + 1}$

$(6\,a)$ $x - 1$	$(6\,b)$ $\begin{cases} x-1, & \text{as / } if\ x \geq 1 \\ 1-x, & \text{as / } if\ x < 1 \end{cases}$	$(6\,c)$ $\pm(x-1)$
$(6\,d)$ geen van hierdie / *none of these*		

Ans: $\sqrt{x^2 - 2x + 1} = \sqrt{(x-1)^2} = |x-1| = \begin{cases} x-1 & \text{as / } if\ x-1 > 0 \\ -(x-1) & \text{as / } if\ x-1 < 0 \end{cases} = \begin{cases} x-1 & \text{as / } if\ x > 1 \\ 1-x & \text{as / } if\ x < 1 \end{cases}$

$(6\,b)$

Vraag 7 / *Question* 7

As / *If* $f(x) = \sin^2 x$, $h(x) = \sin^2\sqrt{x}$ en / *and* $h = f \circ g$, dan is / *then* $g(x) =$

$(7\,a)$ $\sqrt{x}$	$(7\,b)$ $\sin\sqrt{x}$	$(7\,c)$ $\sqrt{\sin x}$	$(7\,d)$ geen van hierdie / *none of these*

Ans: $h(x) = (f \circ g)(x) = f(g(x)) = \sin^2(g(x)) = \sin^2\sqrt{x} \qquad \therefore g(x) = \sqrt{x}$

$(7\,a)$

Vraag 8 / *Question* 8

As / *If* $3e^{\ln x} = 8$ dan is / *then* $x =$

$(8\,a)$ 8	$(8\,b)$ $\frac{8}{\ln 3}$	$(8\,c)$ $\frac{8}{3}$	$(8\,d)$ 2

Ans: $3e^{\ln x} = 8 \Leftrightarrow 3x = 8$ en / *and* $x > 0 \Leftrightarrow x = \frac{8}{3}$.

$(8\,c)$

Vraag 9 / *Question* 9

Bepaal / *Find*: $\lim_{x \to 3} \frac{x^2-9}{x-3}$

(9 *a*) die limiet bestaan nie / *the limit does not exist*	(9 *b*) $\frac{1}{3}$	(9 *c*) ∞	(9 *d*) 6

Ans: $\lim_{x \to 3} \frac{x^2-9}{x-3}$ $\left(\begin{array}{c}\text{Vorm / } \textit{Form} \\ \frac{0}{0}\end{array}\right)$

$= \lim_{x \to 3} \frac{(x-3)(x+3)}{x-3} \lim_{x \to 3} (x+3) = 3+3 = 6$

(9 *d*)

Vraag 10 / *Question* 10

Laat / *Let* $f(x) = \frac{\sqrt{x^2+2}}{x}$

Die horisontale asimptoot(asimptote) van f is / *The horizontal asymptote(s) of* f *is(are)*

(10 *a*) $y = 1$	(10 *b*) $y = -1$	(10 *c*) $y = -1$ en / *and* $y = 1$	(10 *d*) $x = 1$

Ans: $\lim_{x \to -\infty} \frac{\sqrt{x^2+2}}{x}$ $\left(\begin{array}{c}\text{Vorm / } \textit{Form} \\ \frac{\infty}{-\infty}\end{array}\right)$

$= \lim_{x \to -\infty} \frac{\sqrt{x^2}\sqrt{1+\frac{2}{x^2}}}{x}$

$= \lim_{x \to -\infty} \frac{|x|\sqrt{1+\frac{2}{x^2}}}{x}$

$= \lim_{x \to -\infty} \frac{-x\sqrt{1+\frac{2}{x^2}}}{x}$

$= \lim_{x \to -\infty} -\sqrt{1+\frac{2}{x^2}}$

$= -1$

$\lim_{x \to \infty} \frac{\sqrt{x^2+2}}{x}$ $\left(\begin{array}{c}\text{Vorm / } \textit{Form} \\ \frac{\infty}{\infty}\end{array}\right)$

$= \lim_{x \to \infty} \frac{\sqrt{x^2}\sqrt{1+\frac{2}{x^2}}}{x}$

$= \lim_{x \to \infty} \frac{|x|\sqrt{1+\frac{2}{x^2}}}{x}$

$= \lim_{x \to \infty} \frac{x\sqrt{1+\frac{2}{x^2}}}{x}$

$= \lim_{x \to \infty} \sqrt{1+\frac{2}{x^2}}$

$= 1$

$y = -1$ is 'n horisontale asimptoot (links) / $y = -1$ *is a horizontal asymptote (to the left)*
$y = 1$ is 'n horisontale asimptoot (regs) / $y = 1$ *is a horizontal asymptote (to the right)*

(10 *c*)

Vraag 11 / *Question* 11

Bepaal / *Find:* $\lim_{x \to \infty} x\ln x$

(11 *a*) $-\infty$	(11 *b*) 0	(11 *c*) die limiet bestaan nie / *the limit does not exist*	(11 *d*) ∞

Ans: $\lim_{x \to \infty} x\ln x$ $\left(\begin{array}{c}\text{Vorm / } \textit{Form} \\ \infty \times \infty\end{array}\right)$

$= \infty$

(11 *d*)

Vraag 12 / *Question* 12

Laat / *Let* $f(x) = \begin{cases} x^2 - c^2, & \text{as / } if\ x < 4 \\ cx + 20, & \text{as / } if\ x \geq 4 \end{cases}$.

As f kontinu is op $\mathbb{R}$, dan is $c =$ / *If* f *is continuous on* $\mathbb{R}$, *then* $c =$

(12 *a*) ± 2	(12 *b*) -2	(12 *c*) 4
(12 *d*) Dit is onmoontlik om c te vind / *It is impossible to find* c		

Ans: (*i*) $f(4) = 4c + 20$ $\therefore$ f is gedefinieer vir alle waardes van c. / $\therefore$ f *is defined for all values* of c.

(*ii*) $\lim\limits_{x \to 4^-} (x^2 - c^2) = 16 - c^2$ en / *and* $\lim\limits_{x \to 4^+} (cx + 20) = 4c + 20$

$\therefore$ $\lim\limits_{x \to 4} f(x)$ bestaan / *exists* $\Leftrightarrow$ $\lim\limits_{x \to 4^-} f(x) = \lim\limits_{x \to 4^+} f(x) \Leftrightarrow 16 - c^2 = 4c + 20$

$\Leftrightarrow c^2 + 4c + 4 = 0 \Leftrightarrow (c+2)^2 = 0 \Leftrightarrow c = -2$

(*iii*) en / *and* $f(4) = 4c + 20 = 4(-2) + 20 = 12 = \lim\limits_{x \to 4} f(x)$ as / *if* $c = -2$

$\therefore$ Is f kontinu op $\mathbb{R}$, as $c = -2$ / f is *continuous on* $\mathbb{R}$, *when* $c = -2$

(12 *b*)

Vraag 13 / *Question* 13

Bepaal / *Find* $\lim\limits_{x \to \infty} \frac{\sin x}{x}$.

(13 *a*) ∞	(13 *b*) die limiet bestaan nie / *the limit does not exist*	(13 *c*) 0	(13 *d*) 1

Ans: $-1 \leq \sin x \leq 1$ $\therefore$ $\frac{-1}{x} \leq \frac{\sin x}{x} \leq \frac{1}{x}$

$\therefore$ $\lim\limits_{x \to \infty} \left(\frac{-1}{x}\right) \leq \lim\limits_{x \to \infty} \left(\frac{\sin x}{x}\right) \leq \lim\limits_{x \to \infty} \left(\frac{1}{x}\right)$ $\therefore$ $0 \leq \lim\limits_{x \to \infty} \left(\frac{\sin x}{x}\right) \leq 0$

$\therefore$ $\lim\limits_{x \to \infty} \frac{\sin x}{x} = 0$ (knyptangstelling / Squeeze Theorem)

(13 *c*)

[13]

BEANTWOORD ALLE VERDERE VRAE OP HIERDIE VRAESTEL
en TOON alle bewerkings duidelik aan
ANSWER ALL THE FOLLOWING QUESTIONS ON THE SCRIPT
and SHOW all computations clearly.

Vraag 14 / *Question* 14

As $f(x) = \sqrt{x}$, bepaal die waarde (indien dit bestaan) van $\lim\limits_{h \to 0} \frac{f(2+h) - f(2)}{h}$.

If $f(x) = \sqrt{x}$, *find the value (if it exists) of* $\lim\limits_{h \to 0} \frac{f(2+h) - f(2)}{h}$.

Ans: $\lim\limits_{h \to 0} \frac{f(2+h) - f(2)}{h} = \lim\limits_{h \to 0} \frac{\sqrt{2+h} - \sqrt{2}}{h}$ $\left(\text{Vorm/ } \textit{Form} \ \frac{0}{0}\right)$

$= \lim\limits_{h \to 0} \frac{(\sqrt{2+h} - \sqrt{2})}{h} \times \frac{(\sqrt{2+h} + \sqrt{2})}{(\sqrt{2+h} + \sqrt{2})} = \lim\limits_{h \to 0} \frac{(2+h) - 2}{h(\sqrt{2+h} + \sqrt{2})} = \lim\limits_{h \to 0} \frac{h}{h(\sqrt{2+h} + \sqrt{2})}$ $\left(\text{Vorm/ } \textit{Form} \ \frac{0}{0}\right)$

$= \lim\limits_{h \to 0} \frac{1}{(\sqrt{2+h} + \sqrt{2})} = \frac{1}{2\sqrt{2}}$

[3]

Vraag 15 / *Question* 15

Laat / *Let* $f(x) = \tan^{-1} x$ en / *and* $g(x) = |x|$.

(Notasie / *Notation*: $\tan^{-1} x = \arctan x =$ bgtan$x$.)

(15.1) Bepaal / *Find* $(f\circ g)(x)$.

Ans: $(f\circ g)(x) = f(g(x)) = f(|x|) = \tan^{-1}|x| = \begin{cases} \tan^{-1}x, & \text{as / } if\ x \ge 0 \\ \tan^{-1}(-x), & \text{as / } if\ x < 0 \end{cases}$

[1]

(15.2) Stel vas of $f\circ g$ ewe, onewe of nie een van die twee is nie. Verduidelik.
Determine whether $f\circ g$ is even, odd or neither. Explain.

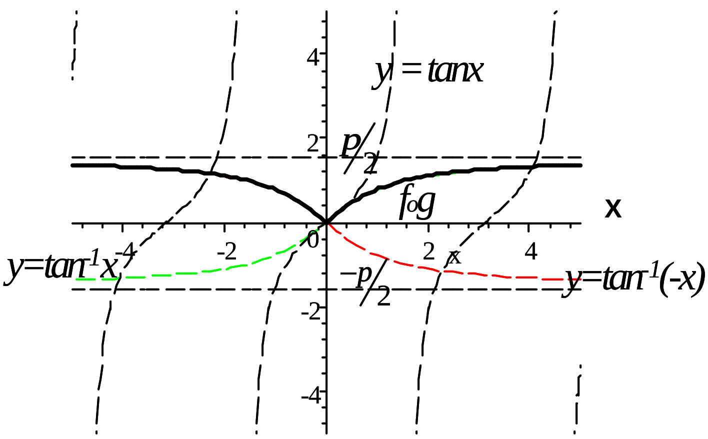
Ans: $(f\circ g)(-x) = f(g(-x)) = f(|-x|) = \tan^{-1}|-x| = \tan^{-1}|x| = (f\circ g)(x)$

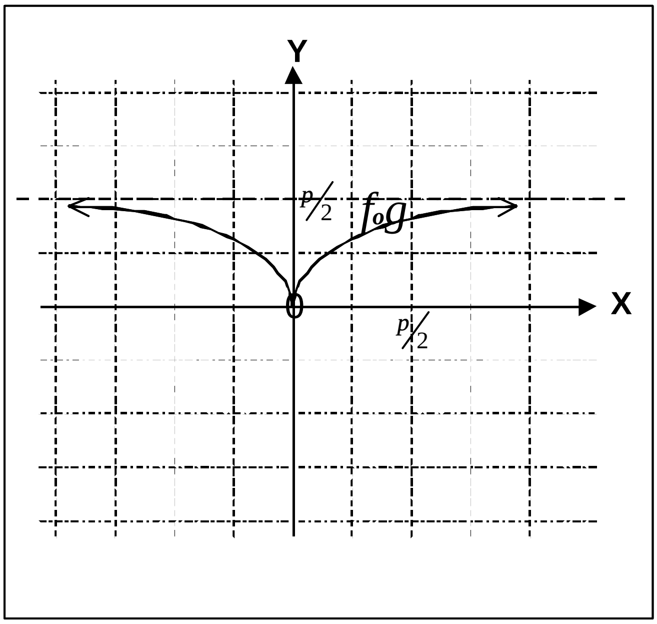
$\therefore f\circ g$ ewe / *even.*

[2]

(15.3) Skets die grafiek van $f\circ g$ op die gegewe assestelsel.
Sketch the graph of $f\circ g$ on the given set of axes.

Ans:





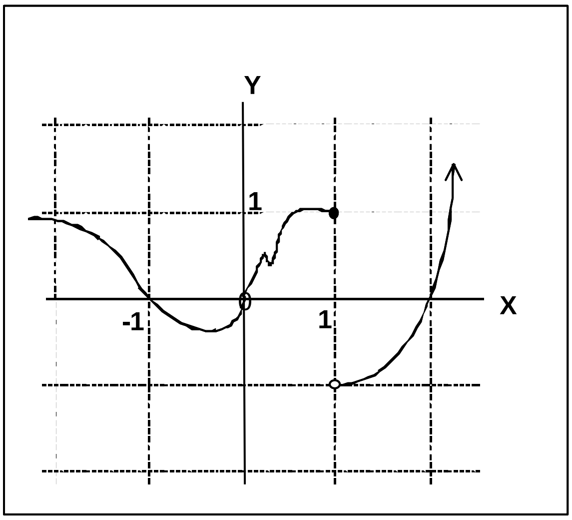
[3]

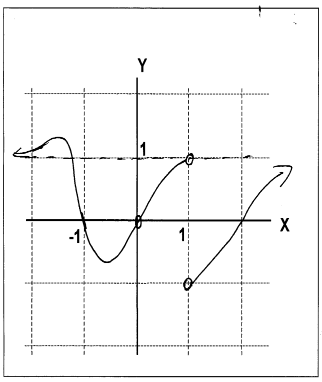
Vraag 16 / *Question* 16

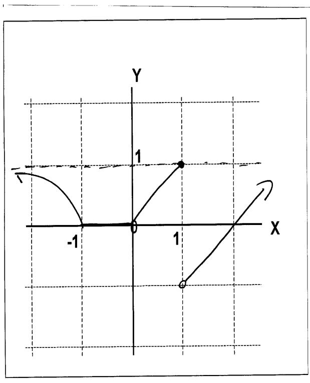
Skets 'n funksie wat die volgende voorwaardes bevredig.
Sketch a function that satisfies the following conditions.

- $f(-1) = f(0) = f(2) = 0$
- $\lim\limits_{x \to 1^-} f(x) = 1$ en / *and* $\lim\limits_{x \to 1^+} f(x) = -1$
- $\lim\limits_{x \to -\infty} f(x) = 1$ en / *and* $\lim\limits_{x \to \infty} f(x) = \infty$.

Ans: Baie moontlikhede / *Many possibilities*







ens, ens, ens. etc. etc, etc

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[3]

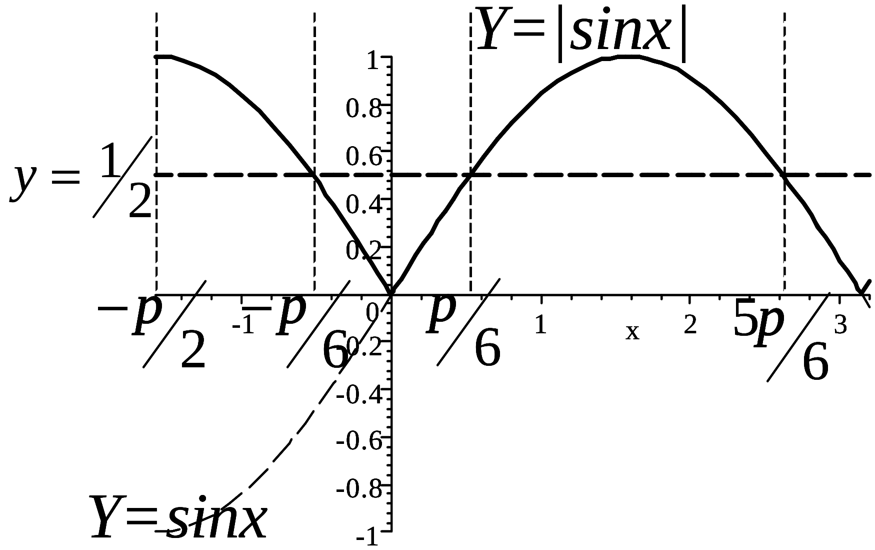
Vraag 17 / *Question* 17

Los op / *Solve* $|2\sin x| \geq 1$ as / *if* $x \in [\frac{-\pi}{2}, \pi]$.

Ans: $|2\sin x| \geq 1 \Leftrightarrow 2|\sin x| \geq 1 \Leftrightarrow |\sin x| \geq \frac{1}{2}$

$\sin x = \frac{1}{2} \Leftrightarrow x = \frac{\pi}{6} + 2k\pi$ of / *or* $x = \pi - \frac{\pi}{6} + 2k\pi, k \in \mathbb{Z}$

$x \in [\frac{-\pi}{2}, \frac{-\pi}{6}] \cup [\frac{\pi}{6}, \frac{5\pi}{6}]$(vanaf grafiek / *from graph*)



[4]

Vraag 18 / *Question* 18

Laat / *Let* $f(x) = \begin{cases} \frac{x^2+2x-3}{x-1} & \text{as / } if\ x \neq 1 \\ a & \text{as / } if\ x = 1 \end{cases}$.

Bepaal die waarde(s) van a sodat f kontinu is in $x = 1$.

Find the value(s) of a such that f is continuous at $x = 1$.

(i) $\quad f(1) = a \quad (\therefore f(1)$ is gedefinieer / *is defined*)

(ii) $\quad \lim_{x \to 1^-} \left(\frac{x^2+2x-3}{x-1} \right) \quad \left(\begin{array}{c} \text{Vorm / } Form \\ \frac{0}{0} \end{array} \right)$

$= \lim_{x \to 1^-} \left(\frac{(x+3)(x-1)}{x-1} \right) = \lim_{x \to 1^-} (x+3) = 4 = \lim_{x \to 1^+} \left(\frac{(x+3)(x-1)}{x-1} \right)$

$\therefore \lim_{x \to 1} \left(\frac{(x+3)(x-1)}{x-1} \right)$ bestaan / *exists*

(iii) Vir f om kontinu te wees moet / *For f to be continuous,*

$\lim_{x \to 1} \left(\frac{(x+3)(x-1)}{x-1} \right) = f(1)$

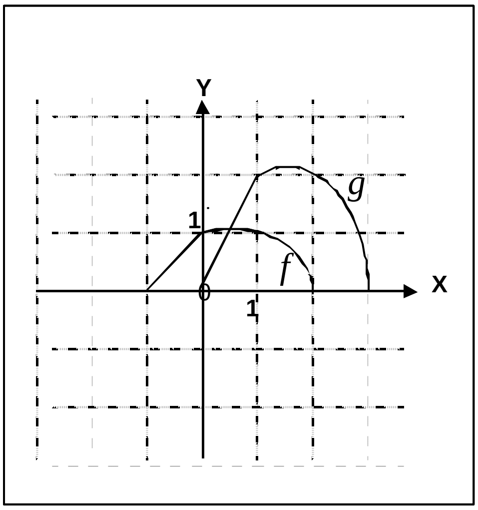
$\therefore a = 4$

[2]

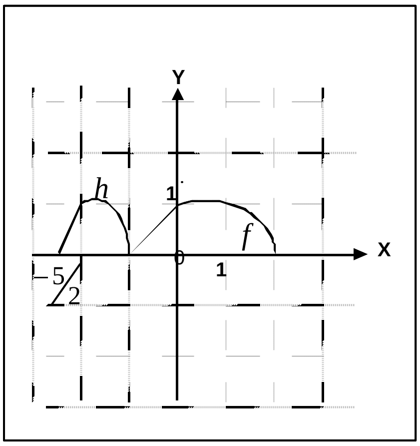
Vraag 19 / *Question* 19

Beskou die grafiek van 'n funksie f.
Consider the graph of the function f.

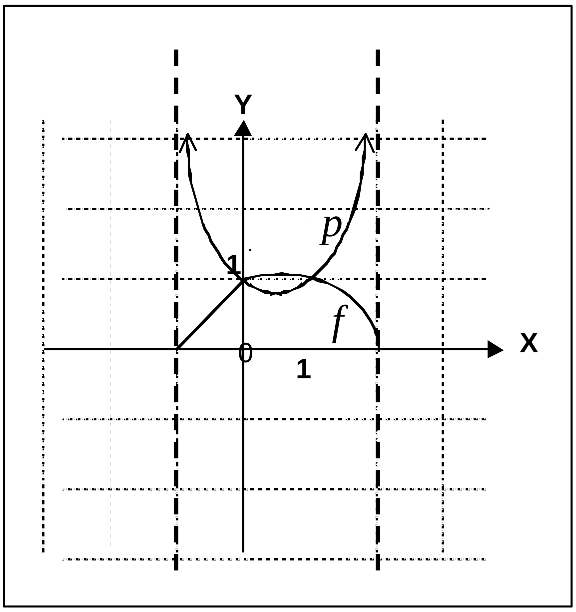
(19.1) Skets 'n nuwe funksie $g(x) = 2f(x-1)$ op diesefde assestelsel as die van f.
Sketch a new function $g(x) = 2f(x-1)$ *on the same set of axes as that of f.*



19.2 Skets 'n nuwe funksie $h(x) = f(2x+4)$ op diesefde assestelsel as die van f.
Sketch a new function $h(x) = f(2x+4)$ *on the same set of axes as that of* f.



19.3 Skets 'n nuwe funksie $p(x) = \frac{1}{f(x)}$ op diesefde assestelsel as die van f.
Sketch a new function $p(x) = \frac{1}{f(x)}$ *on the same set of axes as that of* f.



[4]